

Econometrically Identified Asset-Price Bubbles and Machine-Learning Early-Warning Systems: A Real-Time Financial Econometric Framework

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Abstract

It is difficult to identify asset price bubbles in real time. Many early-warning systems use crisis dates instead, which characterize financial distress after a period of instability has already appeared. In this paper the event to be predicted is the bubble itself. We use the GSADF to identify explosive behavior in prices and, based on the dates of the beginning of the bubble, we determine the target for a machine-learning early-warning model.

The empirical analysis is divided into three parts. First, we implement the GSADF and BSADF tests with wild-bootstrap inference on a set of equity indices and identify 28 dated explosive-price episodes. Second, eight machine-learning models are estimated on a macrofinancial panel of 84 predictors and 311 monthly observations. They use recursive expanding-window estimation to evaluate their performance so that information from later periods is not used to predict earlier ones. Third, we use SHAP values to understand what variables are driving model predictions.

The stacking ensemble achieves an AUROC of 0.967. International equity market variables and monetary conditions with lags are assigned substantial predictive weight. Term spread 12 months lag is ranked 4th in the feature analysis and has an XGBoost SHAP value of 0.727. Time lags are also crucial: 66 of the 84 final features are lagged variables and the Transformer-MLP has a Relative Usefulness of 0.583.

The results demonstrate how econometric bubble dates can be used to establish an early-warning target. These two methods play different roles. The GSADF finds the event, while machine learning uses the information available before the event to estimate the probability of the event. SHAP is then used to explore the variables contributing to those predictions.

Keywords: asset-price bubbles; GSADF; financial econometrics; machine learning; early-warning systems; SHAP; macro financial surveillance; recursive estimation.

The codes could be find at : <https://github.com/Mohsenselseleh/Econometrically-Identified-Asset-Price-Bubbles-and-Machine-Learning-Early-Warning-Systems>

1. Introduction

Asset-price bubbles are hard to detect before they become disruptive. Such standard financial early-warning models predict a crisis based on variables such as credit growth, asset-price growth, leverage, interest rates and financial conditions. This approach results in a measurement problem when the event of interest is the bubble itself and not the financial distress that may follow.

A banking crisis is not the same as an asset-price bubble. A banking crisis is a record of financial distress and institutional fallout. The behavior of the asset price is about a bubble. If the research question is: can explosively price behavior be predicted? then the prediction target should be related to some econometric measure of that behavior.

In this paper we ask whether bubble inception dates from GSADF and BSADF tests can be used to build a machine-learning early-warning system in real time.

The empirical design consists of three parts. First, we detect explosive behavior with the

GSADF and date individual episodes with the BSADF. Second, we convert the inception dates into a binary target. Observations in the four to twelve quarters prior to an identified inception date are treated as pre-bubble observations. Third, we estimate the probability of being in that period using macrofinancial predictors in machine-learning models and study the resulting predictions using SHAP values.

The analysis reveals 28 dated episodes in the investigated equity markets. The recursive out-of-sample evaluation of stacking ensemble leads to an AUROC of 0.967. International equity-market variables and monetary conditions are often among the top predictors, with the 12-month lag of the term spread ranking fourth in the SHAP analysis. The feature set also contains a lot of time information: 66 of the 84 final predictors are lagged variables. The rest of the paper is organized as follows: we first review the relevant literature, then describe the econometric tests, target construction, prediction models and the evaluation procedure. Next we present the empirical results and robustness checks and finally we discuss limitations and future extensions.

2. Related Literature

2.1 Econometric identification of asset-price bubbles

Tests for explosive behavior in financial time series have been widely used in the literature on asset-price bubbles.

The PSY approach to right-tailed recursive unit-root tests developed by Phillips, Shi and Yu. The tests allow the autoregressive root to exceed one in periods of explosive price behavior. GSADF makes this more general by allowing the start of a possible episode to vary. Consequently, many episodes can be studied within the same price series. Subsequently, the individual episodes are dated using the BSADF sequence.

The test asks whether the price series exhibits stronger evidence of explosive behavior than would be expected under the null of no explosive root.

The null hypothesis:

H_0 : The price series does not exhibit explosive behavior.

Hypothesis (alternative):

H_1 : The price series has an explosive section.

The GSADF statistic is the supremum of right-tailed ADF statistics over the admissible windows of samples.

To clarify:

GSADF = maximum right-tailed ADF statistic over the admissible sample windows.

The BSADF sequence is used to identify the relevant critical value of the test statistic.

2.2 Financial early-warning systems

Early warning models employ a large set of macro-economic and financial variables.

Logistic regression and related econometric models are typical baselines, while more recent work uses random forests, support vector machines, gradient boosting, neural networks, and ensemble methods.

These models can allow for nonlinear relationships, interactions, lagged effects, and cross-market links that are difficult to capture with a simple linear specification. They also represent a time series evaluation problem. A random train-test split may contain future information in the training sample and give an overly optimistic view of predictive performance.

Here, the problem is circumvented using recursive expanding-window evaluation.

2.3 The gap between bubble detection and prediction

The literature on econometric bubble detection and the literature on machine-learning early-warning systems have developed mostly independently.

The literature on bubble detection has focused on detecting periods of explosiveness. The early warning literature focuses on predicting crises or financial instability.

The present study links the two.

The logic goes:

Financial prices → GSADF detection → Bubble chronology → Machine-learning prediction → SHAP interpretation.

3. Methodology

3.1 Three-stage empirical approach

The empirical design has three parts.

Stage 1 is bubble detection. GSADF and BSADF tests are used to identify and date explosive episodes.

Stage 2 is prediction. The detected inception dates are converted into a binary pre-bubble target, which is then modeled with the macrofinancial predictors.

Stage 3 is interpretation. SHAP values are calculated for the Random Forest and XGBoost models to examine the variables that contribute to the predictions.

The result is a direct sequence from bubble dating to prediction and then to interpretation.

4. Stage 1: GSADF Bubble Detection

4.1 Data structure

The dissertation applies the GSADF methodology to 11 equity indices and selected financial series. The equity-market analysis produces a database of 28 dated episodes.

The equity indices include the S&P 500, FTSE 100, DAX, CAC 40, Nikkei, KOSPI, BOVESPA, SENSEX, and other international equity indices included in the dissertation.

The minimum window fraction is selected according to the GSADF procedure, and inference is conducted using wild-bootstrap critical values.

4.2 Bootstrap inference

The reported specification uses 99 wild-bootstrap replications. The bootstrap procedure is designed to account for non-standard volatility characteristics in financial data.

The null hypothesis is:

H0: no explosive root exists in the series.

A bubble is considered statistically supported when:
GSADF statistic > bootstrap critical value.

The dissertation also reports a sensitivity analysis using 499 bootstrap replications.

4.3 Main GSADF results

Index	GSADF	5% Critical Value	Reject H0?	Episodes
S&P 500	3.0343	3.2652	No	5
FTSE 100	1.7165	3.0247	No	3
DAX	2.8996	2.8617	Yes	4
CAC 40	1.7049	2.7219	No	4
Nikkei	2.2834	3.1084	No	3
KOSPI	1.9210	3.0554	No	2
BOVESPA	2.1047	3.1218	No	3
SENSEX	1.8832	3.0901	No	2

The full-sample GSADF decision and the BSADF episode dates answer different questions. For the S&P 500, the GSADF statistic is 3.0343 and the 5 percent critical value is 3.2652. The full-sample null is therefore not rejected at the 5 percent level. At the same time, the BSADF procedure dates five separate episodes in the S&P 500 sample. The machine-learning target is based on these dated episodes. It is not based on whether the full sample passes the GSADF rejection threshold.

5. Construction of the Machine-Learning Target

The dependent variable is built from the bubble inception dates.

$Y(t) = 1$ when observation t falls within the defined pre-bubble period before an identified bubble.

$Y(t) = 0$ otherwise.

The main specification uses a four-to-twelve-quarter lead window. In practical terms, the model asks whether information available at a given date contains a signal that the market is approaching a bubble that will be dated later.

The target comes directly from the GSADF/BSADF inception dates. This keeps the event used in the prediction exercise tied to the econometric bubble-detection procedure.

6. Stage 2: Machine-Learning Early-Warning System

6.1 Macrofinancial predictor panel

The machine-learning dataset contains 311 monthly observations and 84 predictors. The variables cover equity-market returns and momentum, volatility, valuation, credit conditions, interest rates, the term structure, international equity markets, and other macrofinancial measures.

Lagged variables make up a large part of the final feature set. Of the 84 predictors, 66 are

lagged. The models can thus use information from earlier months rather than relying only on the current observation.

7. Machine-Learning Models

This analysis evaluates Logistic Regression, Random Forest, XGBoost, Support Vector Machine, k-Nearest Neighbours, Multilayer Perceptron, LSTM-MLP, and Transformer-MLP. The study also evaluates three ensemble specifications: unweighted average, performance-weighted ensemble, and stacking ensemble.

7.1 Logistic regression benchmark

Logistic regression provides the traditional econometric benchmark.

The model estimates the probability of a pre-bubble observation.

In simple form:

Probability of bubble = logistic function of the financial predictors.

The model provides a useful benchmark because its structure is relatively simple and interpretable.

7.2 Random Forest

Random Forest combines many decision trees. Each tree produces a classification decision, and the forest aggregates the results. The objective is to capture nonlinear relationships and interactions between financial variables.

7.3 XGBoost

XGBoost is a gradient-boosting algorithm that sequentially builds decision trees. Each new tree attempts to improve the errors generated by previous trees. This makes the method especially suitable for nonlinear financial relationships.

7.4 LSTM and Transformer models

The LSTM-MLP and Transformer-MLP models are designed to capture temporal information. The sequence models use eight-quarter windows. Instead of observing only the current value of a variable, the models can therefore use a sequence of previous observations. This is important because bubble formation is inherently dynamic.

8. Real-Time Recursive Evaluation

The prediction exercise must avoid future information. For that reason, the models are estimated with an expanding window.

At each step, the model is estimated with information available at time t and produces a prediction for the next period. The new observation is then added to the sample, the model is re-estimated, and the process moves forward. This is repeated over 42 recursive prediction windows.

Training information at time t -> Prediction at time $t+1$.

The design keeps the ordering of information in the evaluation period consistent with a real-time forecasting exercise.

9. Evaluation Measures

AUROC is the main discrimination measure. It summarizes how well a model separates positive and negative observations over different probability thresholds. An AUROC of 0.50 is approximately the level expected from random discrimination.

The analysis also reports the Brier score, Relative Usefulness, the hit rate at a 10 percent false-positive rate, and Model Confidence Set membership.

10. Main Prediction Results

Model	AUROC	Relative Usefulness	Brier Score	Hit Rate @ 10% FPR
Stacking Meta-Learner	0.967	0.333	0.116	0.667
Performance-Weighted	0.809	0.475	0.156	0.545
Unweighted Average	0.798	0.507	0.159	0.455
k-Nearest Neighbours	0.865	0.475	0.144	0.636
Support Vector Machine	0.836	0.443	0.176	0.000
XGBoost	0.815	0.352	0.173	0.545
Random Forest	0.755	0.358	0.165	0.545
Logistic Regression L2	0.610	0.085	0.231	0.182
Logistic Regression L1	0.604	0.170	0.215	0.091
Transformer-MLP	0.694	0.583	0.314	0.000
LSTM-MLP	0.611	-0.083	0.201	0.000

The stacking ensemble records an AUROC of 0.967, a Brier score of 0.116, and a hit rate of 0.667 at a 10 percent false-positive rate.

Performance differs across the individual models. The stacking model combines their predicted probabilities through a meta-learner, so it can use information coming from models with different functional forms.

11. Why the Stacking Model Is Important

The stacking procedure uses the predictions from the individual models as inputs to a meta-learner.

Individual model predictions -> Meta-learner -> Final probability.

The models capture different features of the data. Logistic regression represents relatively simple relationships. Random forests allow nonlinear interactions, XGBoost builds successive improvements to tree-based predictions, kNN uses local similarity, SVM separates observations through classification boundaries, and the neural-network models use more flexible nonlinear functions.

The resulting stacking AUROC is 0.967. This indicates that the combined predictions contain useful information in the recursive evaluation.

12. Temporal Information

The results also show that the timing of the predictors matters. Sixty-six of the 84 final features are lagged variables.

The Transformer-MLP records an AUROC of 0.694 and Relative Usefulness of 0.583. The LSTM-MLP records an AUROC of 0.611 and Relative Usefulness of -0.083.

The sequence models do not uniformly outperform the other models. The results instead indicate that temporal information is useful, while the value of a particular sequence architecture depends on the model and evaluation setting.

13. Stage 3: SHAP Interpretation

Predictive accuracy does not by itself show why a model issues a warning. SHAP values are used for the Random Forest and XGBoost models to examine the contribution of individual variables.

In simple terms:

Final prediction = baseline prediction + contribution of variable 1 + contribution of variable 2 + ...

A positive SHAP contribution raises the predicted probability, while a negative contribution lowers it. The magnitude measures how much a variable contributes to a particular prediction.

These values describe predictive contribution. They do not provide a causal estimate of the effect of a variable on bubble formation.

14. Global Feature Importance

Rank	Feature	RF SHAP	XGB SHAP / Mean
1	DAX_lag12	0.027	0.802 / 0.414
2	DAX_lag6	0.031	0.775 / 0.403
3	FTSE100_lag12	0.014	0.773 / 0.394
4	term_spread_yf_lag12	0.005	0.727 / 0.366
5	term_spread_yf	0.005	0.353 / 0.179
6	US_3M_lag3	0.003	0.339 / 0.171
7	vol_12m_lag6	0.009	0.327 / 0.168
8	KOSPI	0.011	0.287 / 0.149
9	vol_12m_lag3	0.009	0.271 / 0.140
10	Nikkei_lag3	0.003	0.261 / 0.132

International equity-market variables are prominent in the SHAP results. DAX_lag12, DAX_lag6, and FTSE100_lag12 occupy the first three positions in the combined ranking. The 12-month lag of the term spread ranks fourth. Its XGBoost SHAP value is 0.727. Thus, yield-curve information carries substantial predictive weight in the model.

The result should be read as a statement about prediction. It does not establish that changes in the term spread cause bubbles.

15. Economic Interpretation of the Predictors

15.1 International equity markets

DAX, FTSE 100, KOSPI, and Nikkei variables all appear among the predictors. This means that the warning signal is not limited to the domestic market.

Links between markets can arise through portfolio flows, financial conditions, investor sentiment, risk appetite, leverage, exchange rates, and common financial cycles. The role of lagged international equity variables is consistent with this type of cross-market dependence.

15.2 Term structure

The term spread is high in the feature ranking, especially at the 12-month lag.

One possible interpretation is that monetary and financial conditions build over time before speculative price dynamics become detectable. The model, however, does not identify a causal effect of the term spread.

15.3 Volatility

Lagged volatility measures also appear among the leading predictors. This suggests that periods of changing market risk conditions contain information about subsequent bubble dynamics.

The relationship is nonlinear and may depend on the interaction between volatility, monetary conditions, and international financial markets.

16. Historical Episode Interpretation

The SHAP analysis can also be examined episode by episode.

The pre-dot-com and pre-global-financial-crisis housing episodes have relatively high predicted bubble probabilities, but the variables behind the predictions differ. DAX_lag12 receives substantial weight for the pre-dot-com episode, while the lagged term spread is prominent for the pre-GFC housing episode.

This comparison shows why an overall feature ranking does not tell the whole story. A variable can have high average importance while a different variable drives a particular historical prediction.

17. The COVID-19 Episode

The COVID-19 crash offers a useful contrast with the bubble episodes. The model assigns a relatively low bubble probability around that episode.

The result is consistent with a distinction between an endogenous speculative episode and a sudden external shock. The COVID-19 market decline followed a sharp external event rather than the gradual pattern of lagged macrofinancial conditions used in the model.

This also defines the scope of the exercise. The model is aimed at explosive speculative dynamics, so it should not be expected to identify every future market decline.

18. Robustness Analysis

18.1 SMOTE

The results are compared under SMOTE and without SMOTE. The reported AUROC is 0.649 under both configurations in the relevant sensitivity comparison.

This suggests that the class imbalance in the sample does not drive the principal result.

18.2 Prediction horizon

Horizon	AUROC
4-quarter lead	0.497
4-to-8-quarter lead	0.579
4-to-12-quarter lead	0.559

The four-quarter horizon gives an AUROC close to chance. The longer pre-bubble windows perform better.

One possible explanation is that the information relevant to a bubble develops over more than a single quarter. The result also shows that the choice of prediction horizon affects measured performance.

19. Comparison with Crisis-Based Targets

The analysis also compares the GSADF-based target with crisis dates from Laeven and Valencia.

In the relevant US sample, the crisis chronology contains too few positive observations for the same recursive evaluation. Under the time ordering used here, the crisis observation falls into the training period and leaves too few positive cases for the held-out period. The AUROC is consequently undefined for that exercise. GSADF produces multiple dated bubble episodes, which makes it possible to construct a recurring pre-bubble target for the recursive prediction exercise.

20. Econometric Contribution

The econometric contribution is to connect two empirical tasks that are generally treated separately.

The first task is to identify explosive price behavior. The second is to predict whether such behavior is to occur.

The detected inception date is what gives the link between them. We train the machine-learning model on an event defined by the same econometric procedure that identifies the phenomenon of interest, rather than a separate crisis label.

21. Financial-Econometric Interpretation

The results imply a mixture of domestic and international financial conditions during the identified bubble episodes.

The prediction exercise includes international equity markets, interest rates, term spreads,

volatility, lagged market states and cross-market measures. This pattern is consistent with a view of asset-price dynamics in which financial conditions are connected across markets. Machine-learning models can capture nonlinear interactions between these variables but the analysis does not imply a causal interpretation of these relationships.

22. Policy Relevance

The results may be useful for the financial surveillance, as the early-warning model based on the econometrically dated bubbles, adds another source of information to the monitoring process.

A high AUROC alone is not sufficient to justify a model as a policy trigger. The usefulness of a warning depends on the costs of false alarms and missed events, the timing of intervention, data quality, the policy tools available, and the institutional setting. The model is best used as an analytic input to surveillance rather than an automatic decision rule.

23. Limitations

There are some limits:

The reported GSADF specification uses 99 bootstrap replications first. The dissertation also considers 499 and 1,999 replications, which would provide a useful check on the stability of the critical values.

Second, the machine-learning panel does not implement the full covariate-augmented GSADF test; Credit spreads, NFCI, VIX and term-spread variables are used in the prediction stage but only partially in the bubble detection stage.

Third, LPPLS has not been applied for convergent-validity testing.

Fourth, the prediction target is largely concentrated on the US equity-market application, although the bubble-detection stage encompasses a broader set of markets.

Fifth, none of the conventional classifiers were tested with exactly the same recursive design as the LSTM and Transformer models. Thus, comparisons involving these models should be handled with care.

24. Future Research

24.1 Larger bootstrap inference

The GSADF analysis should be repeated using $B = 499$ as the primary bootstrap specification and $B = 1,999$ as a sensitivity analysis.

24.2 Covariate-augmented GSADF

The next extension would put financial covariates directly into the bubble-detection stage. The analysis could then compare:

Univariate GSADF inception date

and Covariate-augmented GSADF inception date.

The difference between the two dates would show whether the added macrofinancial information changes the timing of detection.

24.3 LPPLS validation

LPPLS provides a different approach to identifying accelerating speculative dynamics. Applying LPPLS alongside GSADF would provide a convergent-validity test.

The purpose would not be to replace GSADF but to determine whether the identified episodes are also supported by an alternative bubble-detection methodology.

24.4 Cross-country prediction

The cross-market detection database could eventually be connected to a genuinely international early-warning panel.

Such an approach could investigate whether:

Bubble in Market A → Future bubble probability in Market B.

This would provide a more direct test of financial contagion and cross-border spillovers.

25. Conclusion

The paper is a combination of econometric bubble dating and machine-learning prediction. The episodes are labeled by GSADF and BSADF, the dated inception points establish the pre-bubble target, and the prediction models are based on macrofinancial information available prior to those dates. Then, SHAP values are used to analyze the variables behind the predictions.

Empirical analysis identifies 28 dated episodes. In the recursive out-of-sample exercise, the AUROC achieved by the stacking ensemble was 0.967, its Brier score was 0.116, and its hit rate was 0.667 at a 10 percent false-positive rate.

SHAP results highlight international equity-market variables. The top three are DAX_lag12, DAX_lag6 and FTSE100_lag12, with the 12-month lag of the term spread coming in fourth with an XGBoost SHAP value of 0.727. The feature set also has a lot of time information, 66 out of 84 predictors are lagged variables.

The findings do not suggest that any particular financial variable causes a bubble. They demonstrate that certain market and macrofinancial variables provide predictive information before the dates identified by the GSADF/BSADF procedure.

The study is divided into three questions. When is explosive price behavior found? asks GSADF. The machine-learning models ask if previous information can help identify the pre-bubble period. SHAP asks the question of which variables contribute to those predictions. Future work could consider larger bootstrap samples, covariate augmented GSADF, LPPLS and a broader international prediction panel.

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